

The aim of this project is to develop a **Keyword Extraction Tool** that automatically identifies the most relevant words or key phrases from a collection of text documents.

This system helps summarize large volumes of textual information, making it easier to analyze documents such as **news articles, reviews, or research abstracts**.

Concept Overview

Keyword extraction is a fundamental Natural Language Processing (NLP) task that identifies the most informative terms representing the main themes in a document.

In this project, three different techniques were implemented and compared:

Method	Technique Type	Description
TF-IDF (Term Frequency–Inverse Document Frequency)	Statistical	Measures how important a word is within a document compared to its frequency across all documents.
RAKE (Rapid Automatic Keyword Extraction)	Statistical + Rule-based	Extracts frequent co-occurring words by analyzing word frequency and position, forming meaningful phrases.
KeyBERT	Embedding-based (Deep Learning)	Uses BERT embeddings to extract semantically relevant words/phrases that best represent the document’s meaning.

Implementation Steps

- 1. **Data Collection & Understanding**
 - Used a small, categorized dataset (Business, Tech, Sport, Entertainment, Politics) for demonstration.
 - Each record represented a short text document.
- 2. **Text Preprocessing**
 - Converted text to lowercase.
 - Removed HTML tags, URLs, and punctuation.
 - Tokenized text using `nltk.word_tokenize()`.

- Removed stopwords and applied lemmatization to get root words.
- Result: Cleaned text stored in a new column `clean_text`.

3. Keyword Extraction Techniques

- **TF-IDF**: Extracted top 5 important words from each document.
- **RAKE**: Extracted top-ranked keyword phrases using the RAKE algorithm.
- **KeyBERT**: Used BERT embeddings to extract top 5 semantically relevant keywords.

4. Evaluation

- Compared interpretability, accuracy, and computational speed.
- Verified keyword relevance manually for small samples.
- Observed that **KeyBERT** produced the most meaningful keywords, though slower than TF-IDF and RAKE.

5. Visualization & Reporting

- Generated Word Clouds for most frequent terms.
- Created bar plots to visualize average document lengths per category.
- Constructed a combined **summary table** displaying results from all three methods.

Results

Category	TF-IDF Keywords	RAKE Keywords	KeyBERT Keywords
Business	warner, sale, time, grows, percent	sale boost time warner profit company revenue grows	revenue, profit, percent, warner, sale
Tech	service, spotify, rival, apple, itunes	aol expands music download service rival apple itunes spotify	spotify, aol, itunes, apple, music
Sport	win, thrilling, unbeaten, run, chelseas	arsenal end chelseas unbeaten premier league run thrilling win	arsenal, unbeaten, chelsea, thrilling, win

Category	TF-IDF Keywords	RAKE Keywords	KeyBERT Keywords
Entertainment	weekend, opening,	new hollywood movie break	boxoffice, record,
	record, hollywood, movie	boxoffice record opening weekend	weekend, hollywood, break
Politics	today, prime, reform, parliament, announces	prime minister announces new education policy reform parliament today	reform, parliament, minister, policy, education

Comparative Insights

Metric	TF-IDF	RAKE	KeyBERT
Speed	Fast	Moderate	Slow
Interpretability	Moderate	High	Very High
Semantic Quality	Low (frequency-based)	Medium (phrase-based)	High (context-aware)
Best Use Case	Simple, large datasets	Short texts or reviews	Semantic understanding and summaries

Conclusion

This project demonstrates that combining traditional NLP and modern transformer-based methods can enhance the quality of keyword extraction:

- **TF-IDF** provides fast, explainable insights.
- **RAKE** captures natural language phrase structures.
- **KeyBERT** delivers deep semantic understanding, ideal for real-world applications like summarization and content tagging.

The developed **Keyword Extraction Tool** thus serves as a foundational component for **text analytics, document classification, and automatic summarization** systems.

Technologies Used

- **Python Libraries:** pandas, nltk, sklearn, rake-nltk, keybert, matplotlib, seaborn, wordcloud
- **Environment:** Google Colab
- **Dataset:** Custom BBC-style sample dataset (5 categories)

Future Enhancements

- Integrate with **Hugging Face transformer models** for multilingual keyword extraction.
- Add a **web UI (Gradio or Streamlit)** for real-time keyword extraction.
- Incorporate **automatic evaluation metrics** (Precision, Recall, F1-score) using benchmark datasets.